

Name: ..... Maths Class: .....

# SYDNEY TECHNICAL HIGH SCHOOL



## YEAR 12 HSC COURSE

### Extension 1 Mathematics

#### Assessment 3 June 2013

**TIME ALLOWED: 70 minutes**

***Instructions:***

- *Start each question on a new page.*
- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- It is suggested that you spend no more than 5 minutes on Part A.
- Approved calculators may be used.

**PART A: (5 Marks)**

Answers to these multiple choice should be completed on the multiple choice answer sheet supplied with your answer booklet.

All questions are worth 1 mark

|   |                                                                                                                                                                                                                                    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\frac{d}{dx} \ln\left(\frac{x+1}{2-x}\right) =$<br><br>A. $\frac{3}{(x+1)(2-x)}$<br><br>B. $\frac{1-2x}{(x+1)(2-x)}$<br><br>C. $\frac{1-x}{(x+1)(2-x)}$<br><br>D. $\frac{2x-1}{(x+1)(2-x)}$                                       |
| 2 | An indefinite integral of $\frac{1}{2}(e^x + e^{-x})$ is:<br><br>A. $\frac{1}{2}(e^x + e^{-x})$ B. $-\frac{1}{2}(e^x + e^{-x})$ C. $\frac{1}{2}(e^x - e^{-x})$ D. $-\frac{1}{2}(e^x - e^{-x})$                                     |
| 3 | The indefinite integral of $\frac{1}{\sqrt{9-x^2}}$ is:<br><br>A. $\frac{1}{3}\sin^{-1}\frac{x}{3} + k$ B. $\sin^{-1}\frac{x}{3} + k$ C. $3\sin^{-1}\frac{x}{3} + k$ D. $\frac{1}{3}\sin^{-1}3x + k$                               |
| 4 | The value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ is:<br><br>A. $\frac{\pi}{3}$ B. $\frac{\pi}{6}$ C. $-\frac{\pi}{6}$ D. $-\frac{\pi}{3}$                                                                                  |
| 5 | The curve $y = \frac{1}{\sqrt{16+x^2}}$ between the lines $x = 0$ and $x = 4$ is rotated about the x-axis.<br><br>Its volume is given by:<br><br>A. $\frac{\pi}{16}$ B. $\frac{\pi}{4}$ C. $\frac{\pi^2}{16}$ D. $\frac{\pi^2}{4}$ |

## PART B

### QUESTION 1: (8 marks)

Marks

- (a) Find  $\frac{d}{dx} \tan^{-1} \sqrt{x}$  (simplify your answer) 2
- (b) (i) Find  $\frac{d}{dx} (e^{-x^2})$  1
- (ii) Hence find  $\frac{d^2}{dx^2} (e^{-x^2})$  1
- (c) Evaluate  $\int_0^{\sqrt{2}} \frac{2}{\sqrt{4-x^2}} dx$  2
- (d) If  $y = \tan^{-1} x$  find an expression for  $\sin 2y$ . 2

### QUESTION 2: (Start a new page) (8 marks)

Marks

- (a) Find  $\sin^{-1} \left[ \cos\left(\frac{3\pi}{4}\right) \right]$  1
- (b) You are given the function  $f(x) = (x + 2)^2$
- (i) State the Domain and Range of  $f(x)$  and sketch the curve 2
- (ii) Find the largest possible domain of  $f(x)$ , containing the point  $(0, 4)$  for an inverse function  $y = f^{-1}(x)$  to exist. 1
- (iii) Find the inverse function  $y = f^{-1}(x)$  from part (ii) above and give its Domain and Range 3
- (iv) Sketch  $y = f^{-1}(x)$  1

**QUESTION 3: (Start a new page) (8 marks)**

**Marks**

- (a) Find the exact area beneath the curve  $y = \frac{e^x}{1+e^x}$  above the  $x$ -axis, and between the lines  $x = 0$  and  $x = 1$  2
- Give your answer in simplest terms.
- (b) A radio-active substance decomposes, and the mass present ( $M$ ) after  $t$  years from a certain date is given by  $M = M_0 e^{-kt}$  where  $M_0$  and  $k$  are constants
- (i) Show that this is a solution to  $\frac{dM}{dt} = -kM$  1
- (ii) If the initial mass is 100 gm and the mass after 2 years is 80 gm, find the value of  $k$  to 2 dec. places. 2
- (iii) Find the number of years taken for the mass to halve (called the *half life* of the substance). Give your answer to 1 decimal place. 2
- (iv) Sketch the graph of  $M = M_0 e^{-kt}$  using the vertical axis as  $M$  and the horizontal axis as  $t$ . *Show only keypoints.* 1

**QUESTION 4: (Start a new page) (8 marks)**

**Marks**

- (a) (i) Find  $\frac{d}{dx}(\sin^{-1}x + \cos^{-1}x)$  1
- (ii) Hence find the exact value of  $\sin^{-1}x + \cos^{-1}x$  2
- You must justify your answer NOT just state it.
- (b) Evaluate  $\int 2^x dx$  1
- (c) The pressure  $P \text{ gm/cm}^3$  on a mass of gas, of volume  $v \text{ cm}^3$  is given by the formula 4
- $Pv = 1500.$
- If the volume is increasing at the rate of  $10 \text{ cm}^3/\text{sec}$ , find the rate at which the pressure is decreasing when the volume is  $30 \text{ cm}^3$

**QUESTION 5 : (Start a new page) (8 marks)**

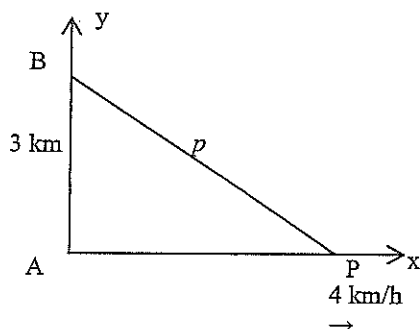
**Marks**

(a) (i) Find  $\frac{d}{dx}(\ln(\sin x))$  1

(ii) Hence, or otherwise, find  $\int \cot 3x \, dx$  1

(b) Find  $\int \frac{dx}{9+4x^2}$  2

(c) A person, P, is walking directly east from a point A at a speed of 4 km/h (ie  $\frac{dx}{dt} = 4$ ) and is being watched by an observer at a point B, which is 3 km due north of A, as shown below: 4



The distance between the observer and the walker is given as  $p$  km.

Find the rate of change of  $p$  when P has walked 4 km?

**QUESTION 6: (Start a new page) (8 marks)**

**Marks**

- (a) Show that the curve  $y = \frac{e^{-x}}{1+x^2}$  is decreasing for all  $x$ , except  $x = -1$  **3**
- (b) The population of seals on an island is increasing at a variable rate, and the number of seals ( $P$ ) at any time  $t$ , is given by
- $$P = A(1 - e^{-kt}), \text{ where } A \text{ and } k \text{ are constants}$$
- (i) Show that  $\frac{dP}{dt} = k(A - P)$  **1**
- (ii) Show that the maximum seal Population that the island can accommodate is  $A$ . **1**
- (iii) If one quarter of the maximum population that the island can hold is reached after 5 hours, what fraction is populated after another 5 hours? **3**

***End of Examination***

# SOLUTIONS

## PART A

- 1/ A      2/ C      3/ B      4/ C      5/ C

## PART B:

### QUESTION 1:

(a)  $\frac{\frac{1}{2}x^{-1/2}}{1+x} = \frac{1}{2\sqrt{x}(1+x)}$

(b) (i)  $\frac{dy}{dx} = -2xe^{-x^2}$

(ii)  $\frac{d^2y}{dx^2} = e^{-x^2}(-2) + (-2x)(-2x)e^{-x^2}$   
 $= -2e^{-x^2}[1-2x^2]$

1 for the differential  
1 for simplified answer

1 MARK

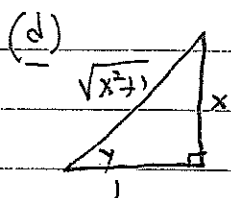
1 MARK

(or  $2xe^{-x^2}(2x^2-1)$ )

(c)  $2\sin^{-1} \frac{x}{2} \Big|_0^{\sqrt{2}} = 2\sin^{-1} \left(\frac{1}{\sqrt{2}}\right) - 2\sin^{-1} 0$   
 $= 2\pi/4$   
 $= \pi/2$

1 for the integral

1 for  $\pi/2$



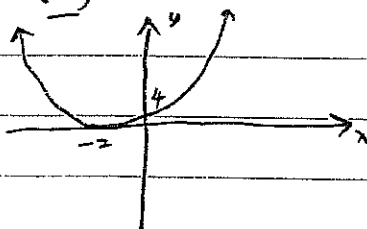
$\sin 2y = 2\sin y \cos y$   
 $= 2 \frac{x}{\sqrt{x^2+1}} \cdot \frac{1}{\sqrt{x^2+1}}$   
 $= \frac{2x}{x^2+1}$

1 for each of  
 $\sin y$   
and  $\cos y$

### QUESTION 2:

(a)  $-\pi/4$

(b)  $\mathcal{D}$ : all  $x$      $\mathcal{R}$ :  $y \geq 0$



1 MARK

1 for both  $\mathcal{D}$  and  $\mathcal{R}$ .

1 for the sketch

(ii)  $\mathcal{D}_f: x \geq -2$  for  $f^{-1}$  to exist

← ①

(iii)  $y = (x+2)^2 \rightarrow x = (y+2)^2$   
 $y = \sqrt{x} - 2$

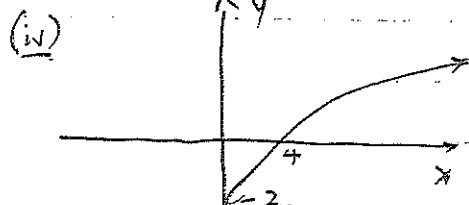
1 MARK for  $f^{-1}(x)$

$\therefore f^{-1}(x) = \sqrt{x} - 2$      $\mathcal{D}_{f^{-1}}: x \geq 0$

1 MARK

$\mathcal{R}_{f^{-1}}: y \geq -2$

1 MARK



1 MARK

QUESTION 3:

(a)  $\int_0^1 \frac{e^x}{1+e^x} dx = \ln(1+e^x) \Big|_0^1$   
 $= \ln(1+e) - \ln(2)$   
 $= \ln\left[\frac{1+e}{2}\right]$

1 for  $\ln(1+e^x)$

1 for either

(b) (i)

$$\frac{dM}{dt} = -kM_0 e^{-kt}$$
$$= -kM$$

1 MARK

(ii)  $M_0 = 100 \Rightarrow M = 100e^{-kt}$

At  $t=2$   $M=80$

$$\therefore 80 = 100e^{-2k}$$

$$\therefore -2k = \ln 0.8$$

$$\therefore k \approx 0.11$$

2 MARKS

(iii) At  $M=50$

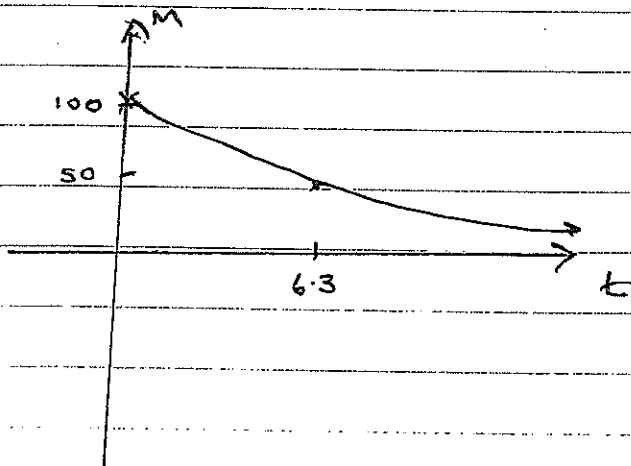
$$50 = 100e^{-0.11t}$$

$$\therefore -0.11t = \ln 0.5$$

$$t \approx 6.3 \text{ years}$$

2 MARKS

(iv)



1 for a "decent" graph

Question 4:

(a) (i)  $\frac{1}{\sqrt{1+x^2}} + \frac{-1}{\sqrt{1-x^2}} = 0$

(ii)  $\therefore \sin^{-1}x + \cos^{-1}x = c$  a constant  
 Since  $\frac{d}{dx}(c) = 0$

At  $x=0$   $c = \sin^{-1}(0) + \cos^{-1}(0)$   
 $= \pi/2$

$\therefore \sin^{-1}x + \cos^{-1}x = \pi/2$

← ①

2 MARKS

1 for finding  $\pi/2$   
 and 1 for a reasonable justification of why.

(b)  $\frac{1}{\ln} 2^x$

①

(c)  $P = \frac{1500}{v}$

$\therefore \frac{dP}{dv} = -1500/v^2$  and  $\frac{dv}{dt} = 10$

$\therefore \frac{dP}{dt} = \frac{dP}{dv} \times \frac{dv}{dt}$

$= -1500/v^2 \times \frac{10}{1}$

$= -15000/900$

$= -50/3 \text{ gm/cm}^3/\text{sec}$

1 MARK for each  
 of  $\frac{dv}{dt}$  and  $\frac{dP}{dv}$ .

← 1 to get here

1 MARK

QUESTION 5:

(a) (i)  $\frac{\cos x}{\sin x} = \cot x$

(ii)  $\int \cot 3x dx = \frac{1}{3} \ln(\sin 3x) + k$

(b)  $\frac{1}{4} \int \frac{dx}{9/4 + x^2} = \frac{2}{3} \cdot \frac{1}{4} \tan^{-1} \frac{2x}{3} + k$   
 $= \frac{1}{6} \tan^{-1} \frac{2x}{3} + k$

2 MARKS

1 only for missing the  $\frac{1}{6}$

(c)  $\frac{dn}{dt} = 4$   $p = \sqrt{9+x^2}$

$\frac{dp}{dx} = \frac{1}{2} \cdot 2x(9+x^2)^{-1/2}$

$\frac{dp}{dt} = \frac{dp}{dx} \cdot \frac{dx}{dt}$

$= \frac{4x}{\sqrt{9+x^2}}$

← 1 for P.

← 1 for  $\frac{dp}{dx}$

← 1 for  $\frac{dp}{dt}$

At  $x=4$   $\frac{dp}{dt} = \frac{16}{5} \text{ km/hr}$

← ①

QUESTION 6:

$$(a) \frac{dy}{dx} = \frac{-(1+x^2)e^{-x} - e^{-x} \cdot 2x}{(1+x^2)^2}$$

$$= \frac{-e^{-x} [1+x^2+2x]}{(1+x^2)^2}$$

$$= \frac{-e^{-x} (1+x)^2}{(1+x^2)^2} \neq 0$$

$$\text{because } (1+x^2) \geq 0$$

$$(1+x^2)^2 \geq 0$$

$$e^{-x} > 0$$

$\forall x$  values

$$\text{when } x = -1 \quad \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} < 0 \quad \forall x \text{ except } x = -1$$

$\therefore$  decreasing  $\forall x$  " " " "

(b)

$$\begin{aligned} (i) \quad \frac{dP}{dt} &= -Ae^{-kt}(-k) \\ &= Ake^{-kt} \\ &= k(A - A + Ae^{-kt}) \\ &= k[A - A(1+e^{-kt})] \\ &= k(A - P) \end{aligned}$$

$$(ii) \quad \text{As } t \rightarrow \infty, e^{-kt} \rightarrow 0, \text{ so } P \rightarrow A$$

This means the population reaches a capacity of A (or the island is "full")

$$(iii) \quad \text{At } t=5, P = \frac{1}{4}A$$

$$\therefore 0.25 = 1 - e^{-5k}$$

$$\therefore 5k = \ln(0.75)$$

$$k = 0.0575$$

$$\text{At } t=10 \quad \frac{P}{A} = 1 - e^{-0.575}$$

$$= 0.4372$$

$$\approx 44\%$$

1 mark.

1 for stating these, or equivalent statements

1 for realising  $x = -1$  gives the zero.

1 for any reasonable attempt

← ①

← ①

① for k.

① for 44%, 43.7% or equivalent.